

Project Executable Files

1. Introduction

The OptiCrop project consists of multiple executable and supporting files that collectively implement an intelligent crop recommendation system. These files are organized to facilitate machine learning model execution, web application deployment, data preprocessing, visualization, and project documentation. Proper organization of executable files improves maintainability, scalability, and ease of deployment.

2. Repository Overview

The project repository contains source code, trained machine learning models, datasets, notebooks, frontend resources, and documentation required for successful execution of the application.

3. Executable Files

The following files are responsible for executing different components of the project.

File Name	Purpose
app.py	Main Flask application responsible for handling requests, processing user inputs, and generating crop recommendations.
model.pkl	Stores the trained Logistic Regression model used for prediction.
scaler.pkl	Stores the trained StandardScaler used for feature scaling.
label_encoder.pkl	Converts predicted numerical labels into crop names.
requirements.txt	Contains all Python dependencies required to run the project.

OptiCrop_Model.ipynb Complete notebook containing data preprocessing, exploratory analysis, model development, evaluation, and model saving.

preprocessing.ipynb Performs dataset cleaning and preprocessing.

visualization.ipynb Generates visualizations and exploratory data analysis.

These files correspond to the implementation included in the project repository.

4. Project Directory Structure

The project is organized into multiple directories for better maintainability.

Folder	Description
Dataset	Contains the Crop Recommendation dataset used for model training.
Flask	Contains the web application source code and trained model files.
Templates	HTML pages used for the web interface.
Static	CSS files used for application styling.
Data Preprocessing	Notebook for cleaning and preparing the dataset.
Visualization Analysis	Notebook containing exploratory data analysis and visualizations.
Notebook	Complete machine learning workflow.

Documentation Project reports and supporting documents.

5. Software Requirements

The project requires the following software components for execution.

Software	Purpose
Python 3.x	Application development
Flask	Web framework
Scikit-learn	Machine learning
Pandas	Data processing
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Exploratory data analysis
Git	Version control
Jupyter Notebook	Model development

The required Python libraries are listed in the project's dependency file.

6. Steps to Execute the Project

The application can be executed using the following procedure:

1. Clone the project repository.
2. Navigate to the project directory.
3. Install all required dependencies using the `requirements.txt` file.
4. Run the Flask application.
5. Open the application in a web browser.
6. Enter the required soil and environmental parameters.
7. Submit the form to receive the recommended crop.

These execution steps are consistent with the existing project instructions.

7. Expected Output

After successful execution, the application allows users to:

- Enter agricultural parameters.
- Generate crop recommendations.
- View prediction results through a web interface.
- Navigate between different application pages.

8. Advantages of the Project Structure

The current project organization provides several benefits:

- Modular implementation.
- Easy maintenance.
- Simple deployment.
- Clear separation of source code and documentation.
- Efficient management of trained machine learning models.
- Improved readability for future development.

9. Conclusion

The executable files and organized project structure ensure that the OptiCrop application can be developed, maintained, and executed efficiently. By separating the machine learning components, web application, datasets, and documentation into dedicated modules, the project remains scalable, maintainable, and easy to understand. This structured organization supports smooth deployment and future enhancements while enabling reliable crop recommendation functionality.